

1. Problem 1 (4.10)

$$E = J(J+1) \frac{\hbar^2}{2I}$$

$$I = \mu R^2$$

$$\frac{I}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$$

$$\mu_m = \frac{m_1 m_2}{m_1 + m_2}$$

$$I = \frac{(14 \cdot 10^{-3})^2}{28 \cdot 10^{-3}} \frac{1}{6.022 \cdot 10^{23}} (1.098 \cdot 10^{-10})$$

$$I = 1.401 \cdot 10^{-46}$$

$$J(J+1) = 4(4+1) = 20$$

$$E = \frac{20 (6.625 \cdot 10^{-34})^2}{4\pi^2 \cdot 2 \cdot 1.401 \cdot 10^{-46}}$$

$$E = 7.94 \cdot 10^{-22}$$

$$\frac{3}{2} k_B T = \frac{3}{2} \cdot 1.38 \cdot 10^{-23} \cdot 400$$

$$\frac{3}{2} k_B T = 8.828 \cdot 10^{-21}$$

$$\frac{E}{\frac{3}{2} k_B T} = \frac{7.94 \cdot 10^{-22}}{8.828 \cdot 10^{-21}}$$

$$\boxed{\frac{E}{\frac{3}{2} k_B T} = 0.096}$$

2. Problem 2 (4.11)

a.

$$\Delta E = h\nu$$

$$\Delta E = hc\tilde{\nu}$$

$$E = J(J+1)\frac{\hbar^2}{2I}$$

$$\Delta E = \frac{\hbar^2}{2I} [J_1(J_1+1) - J_2(J_2+1)]$$

$$\hbar = \frac{h}{2\pi}$$

$$hc\tilde{\nu} = \frac{\left(\frac{h}{2\pi}\right)^2}{2I} [J_1(J_1+1) - J_2(J_2+1)]$$

$$J_1 = 2$$

$$J_2 = 1$$

$$hc\tilde{\nu} = \frac{h^2}{8\pi^2 I} [2(2+1) - 1(1+1)]$$

$$I = \frac{h}{2\pi c\tilde{\nu}}$$

$$\nu = 7.71 \text{ cm}^{-1}$$

$$c = 3 \cdot 10^{10}$$

$$I = \frac{6.626 \cdot 10^{-34}}{2\pi \cdot 3 \cdot 10^{10} \cdot 7.71}$$

$$I = 1.451 \cdot 10^{-46} \text{ kg}\cdot\text{m}^2$$

b.

CO reduced mass

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$\mu = \frac{12 \cdot 16}{12 + 16} \cdot \frac{1}{6.022 \cdot 10^{23}} \cdot \frac{1}{1000}$$

$$\mu = 1.139 \cdot 10^{-26}$$

Reduced mass from inertial moment

$$I = \mu R^2$$

$$\mu = \frac{I}{R^2}$$

$$R = 1.1281 \cdot 10^{-10}$$

$$\mu = \frac{1.451 \cdot 10^{-46}}{(1.1281 \cdot 10^{-10})^2}$$

$$\mu = 1.140 \cdot 10^{-26}$$

The two masses are approximately the same. This molecule is $^{12}\text{C}^{16}\text{O}$.

3. Problem 3

- First R branch peak corresponds to V,J (0,0) $\rightarrow$ (1,1).
- The missing peak is the Q-branch which is J=0 which is not allowed.
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$$\begin{aligned}
 4B &= 2321 - 2295 = 26 \\
 B &= \frac{26}{4} = \frac{6.5}{5.034 \cdot 10^{22}} = 1.29 \cdot 10^{-22} \text{ J} \\
 B &= \frac{\hbar^2}{2I} \\
 I &= \frac{\hbar^2}{2B} \\
 I &= \frac{(1.04 \cdot 10^{-34})^2}{2 \cdot 1.29 \cdot 10^{-22}} = 4.27 \cdot 10^{-47} \\
 \mu &= \frac{1 \cdot 127}{1 + 127} = \frac{0.9922}{6.022 \cdot 10^{26}} \\
 \mu &= 1.648 \cdot 10^{-27} \text{ kg} \\
 I &= \mu R^2 \\
 R &= \sqrt{\frac{I}{\mu}} \\
 R &= \sqrt{\frac{4.27 \cdot 10^{-47}}{1.648 \cdot 10^{-27}}} \\
 \boxed{R = 1.61 \cdot 10^{-10} \text{ m}}
 \end{aligned}$$

- Q branch line is halfway between the peaks 2295 cm^{-1} and 2321 cm^{-1} .

$$\begin{aligned}
 \tilde{\nu} &= 2308 \text{ cm}^{-1} \\
 \nu &= c\tilde{\nu} \\
 \omega &= 2\pi\nu \\
 \omega &= 2\pi c\tilde{\nu} \\
 \omega &= \sqrt{\frac{k}{\mu}} \\
 k &= \omega^2 \mu \\
 k &= (2\pi c\tilde{\nu})^2 \mu \\
 k &= (2\pi \cdot 3 \cdot 10^{10} \cdot 2308)^2 \cdot 1.648 \cdot 10^{-27} \\
 \boxed{k = 311.9 \text{ N/m}}
 \end{aligned}$$

- Replacing H with D would increase the reduced mass which would decrease B and decrease the vibrational frequency. Decreasing B would decrease the space between peaks. Decreasing the vibrational frequency would shift the wavenumbers of the whole spectrum down.

4. Problem 4

$$\Delta E = \Delta n^2 \frac{h^2}{8mL^2}$$

$$\lambda = \frac{hc}{\Delta E}$$

$$\lambda = \frac{hc}{\Delta n^2 \frac{h^2}{8mL^2}} = \frac{8cmL^2}{\Delta n^2 h}$$

$$\lambda = \frac{8 \cdot 3 \cdot 10^8 \cdot 9.11 \cdot 10^{-31} \cdot 2.64 \cdot 10^{-9}}{(12^2 - 11^2) \cdot 6.626 \cdot 10^{-34}}$$

$$\lambda = 1 \cdot 10^{-6} \text{ m}$$

$$\boxed{\lambda = 1000 \text{ nm}}$$

$$\boxed{\lambda = 1 \text{ } \mu\text{m}}$$